

LEARNING OUTCOMES!

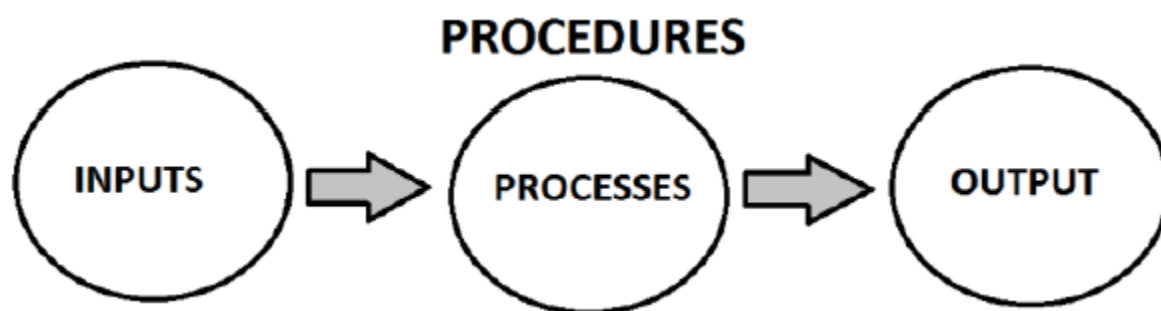
Candidates should be able to:

- understand how small-scale subsistence farming, cash crop farming and livestock farming operate as systems made up of inputs, processes and outputs.
- identify on a map the main areas where cotton, rice, sugar cane and wheat are grown, and the main areas where buffalo, cattle, goats, sheep and poultry are reared
- recognise (from photographs) fields of cotton, rice, sugar cane and wheat; recognise (from photographs) buffalo, cattle, goats, sheep and poultry
- state the uses of the crops named above
- state the main products of the livestock named above and the uses of those products
- identify the main areas for the cultivation and growth of each of the following: apples, apricots, bananas, dates, maize, mangoes, millet, oilseeds, oranges, pulses, tobacco and vegetables. Know why they are grown there and state an important use of each.
- explain how natural and human factors affect production on small-scale subsistence farms, including: - rice grown using traditional methods of ploughing, transplanting, irrigating, harvesting and threshing on small, fragmented holdings using family labour - wheat grown in areas dependent upon rainfall (barani farming areas) - dates and vegetables grown using karez irrigation in a desert oasis
- explain how natural factors, including climatic requirements, and human factors affect the production of cotton, rice, sugar cane (kharif crops) and of wheat (a rabi crop) under the cash crop farming system
- explain how natural and human factors affect livestock farming (poultry farming, the keeping of buffalo and cattle, the keeping of livestock) on small-scale subsistence farms and the keeping of cattle, goats and sheep on a nomadic or semi-nomadic basis, including transhumance
- understand how government action has helped to increase production through land reforms, the promotion of training and the use of machinery, chemicals, improved seeds and other means

- understand and evaluate the possibilities for and problems of the development of agriculture and its sustainability.

AGRICULTURE

- ✓ Agriculture is a combination of two words i.e. Agri and culture. Agri means farming and culture means practice, it means practice of farming is called agriculture.
- ✓ Agriculture is the process by which food crops and other goods are produce, including Wheat, Rice, Sugarcane, Maize along with Cotton etc.



SUBSISTENCE FARMING

- ✓ Subsistence farming means growing of crops mainly for the people who work on them.
- ✓ Surplus is sold in the local village market.
- ✓ Most of the farmers have to supplement their income from other sources e.g. carpenter, blacksmith, cobbler.

INPUTS	PROCESSES	OUTPUTS
Land	Ploughing by wooden plough	Rice
Soil	Sowing by Labor (Desi Seeds)	Wheat
Climate	Irrigation by Traditional methods	Maize
Water	Fertilizing by Natural Manure	Vegetables
Natural Manure	Weeding by Labor	
Draft Power	Threshing by Labor	
Desi Seeds	Harvesting by Labor	
Traditional Plough		
Labour		
Inherited Knowledge		

Why does the output of a small-scale subsistence farm vary from year to year?

- ✓ Variable rainfall
- ✓ Flooding
- ✓ Traditional methods of irrigation used which are unreliable
- ✓ Pests and diseases attack as pesticides not used by the farmer
- ✓ Uses own seed (Desi) / not HYV

CASH CROP FARMING

- ✓ Cash crop farming means the growing of crops for sale
- ✓ A crop that is grown primarily for sale is called a cash crop

INPUTS	PROCESSES	OUTPUTS
Land	Ploughing by Tractors	Rice
Soil	Sowing by Machines (HYV)	Wheat
Climate	Irrigation by Modern Methods	Maize
Water	Fertilizing by Chemical Fertilizers	Vegetables
Fertilizers	Weeding by Labor	Cotton
Machinery	Threshing by Thresher	Sugarcane
HYV seeds	Harvesting by Harvester	
Pesticides		
Skilled Labour		
Knowledge		

Read the following two views:

A

Pakistan should plant more cash crops on its land to generate export earnings.

B

More land should be used to grow crops to feed the growing population of Pakistan.

Which view do you agree with more? Give reasons to support your answer and refer to examples you have studied.

I agree with view B more.

Why?

- ✓ Population growing rapidly (1.6% per annum)
- ✓ Increasing demand for food
- ✓ Fertile land becoming scarce (due to waterlogging and salinity / desertification / soil erosion / over cultivation)
- ✓ Saves expensive imports of food
- ✓ Can be grown on subsistence farms / at low cost (using traditional methods / implements / family labour / small holdings)
- ✓ Many farmers cannot afford cost of modern agricultural methods in cash crop farming
- ✓ Cash crops are monocultures vulnerable to disease.
- ✓ Uses chemical inputs such as fertilisers / pesticides which can pollute water

NATURAL INPUTS

- ✓ Agricultural requirements provided by nature e.g. Land, Soil, Climate.

FLAT LAND

- ✓ Use of machines.
- ✓ Crops grow at equal height, easy to harvest
- ✓ Distribution of water equally
- ✓ Less soil erosion
- ✓ Less drainage
- ✓ Easy to cultivate
- ✓ Provision of canal irrigation

Explain how farm size can affect production on farms in Pakistan.

- ✓ Higher proportion wastage in smaller farms
- ✓ Difficult to use machinery on smaller farms
- ✓ Less able to grow crops for sale as output is low in smaller farms
- ✓ Less able to produce quality crops
- ✓ Lower yields (as cannot afford / do not have modern irrigation / tubewells)

SOIL

- ✓ Soil is the material in which the roots of the plant are embedded.
- ✓ A soil suitable for growth contains sufficient minerals for crop growth and also has sufficient pore spacing.
- ✓ Pore spacing is important as it controls the amount of air and water available for plant roots.
- ✓ Too little spacing (clayey) soil means that it will contain less air and more water, thus won't support plant growth
- ✓ On the contrary a sandy soil will allow air but will also allow a lot of water to infiltrate into the subsoil. This water may also take down with it fertilizers and other natural minerals (which are wasted as roots did not have time to absorb them)
- ✓ The best soil is loamy soil, which contains sufficient pore spaces (to allow for sufficient air and moisture). Spaces aren't too big; so the soil does retain nutrients, which are not leached into the soil
- ✓ The soil must be deep and must contain nitrates and phosphates; which are required for crop growth

What is alluvial soil?

- ✓ silt/loam/sediment deposited by rivers from flooding
- ✓ contains nutrients/minerals

Explain why alluvial soil is good for crop growth.

- ✓ Fertile/contains nutrients (e.g. nitrate/potash/phosphate)
- ✓ fine texture for drainage/not prone to waterlogging
- ✓ retains moisture/moisture retentive
- ✓ replaced each year

CLIMATE

- ✓ Every crop has its own distinct climatic requirements. This includes rainfall, humidity, temperature, amount of sunlight etc.
- ✓ In Pakistan, these requirements basically fall into two groups, Kharif and Rabi.

Sunshine

- ✓ Photosynthesis
- ✓ Warmth
- ✓ Ripening

Temperature

- ✓ For growing

Rainfall

- ✓ For germination
- ✓ Growing
- ✓ Swelling the grains
- ✓ Soften the soil for ploughing \ sowing
- ✓ Planting seeds
- ✓ For flooding fields for rice

DISADVANTAGES OF NATURAL INPUTS

Early Arrival

- ✓ Low yield and income
- ✓ Flooded or water logged
- ✓ Seedlings may be washed away
- ✓ Ground too wet
- ✓ Encourages pests and diseases

Late Arrival

- ✓ Low yield and income
- ✓ May delay planting or sowing
- ✓ Growth stops
- ✓ Irrigation, costly
- ✓ Shortage of water

Variation Of Rainfall

- ✓ Irrigation is costly
- ✓ Shortage of water
- ✓ Unreliable income
- ✓ Livestock may die

Thunderstorms

- ✓ Damage crop
- ✓ Soil erosion

Frosts

- ✓ Kill plants
- ✓ Growth is reduced

Droughts

- ✓ Kill young plants
- ✓ Seeds can die
- ✓ Growth is reduced

Floods

- ✓ Wash the crops away
- ✓ Soil erosion

HUMAN INPUTS

- ✓ Agricultural requirements provided by man. e.g. Capital, Labor.

CAPITAL

- ✓ This is all the money and machinery that is invested into the farm including tractor and harvester etc

Money

- ✓ To buy seeds
- ✓ To buy Chemical fertilizers
- ✓ To buy HYV's
- ✓ To buy Machines
- ✓ To buy Land
- ✓ To pay Labor
- ✓ Irrigation

Advantages Of Machinery

- ✓ Less time consuming
- ✓ Less labor
- ✓ Better work
- ✓ Better processes
- ✓ No need of rest

Disadvantages Of Machinery

- ✓ Expensive to buy or import
- ✓ High maintenance cost
- ✓ Difficult to repair
- ✓ Unemployment
- ✓ Skilled labor required
- ✓ Cannot be used in mountains
- ✓ Cannot be used in small lands

Why are tractors not used by many small-scale farmers?

- ✓ Too expensive to buy

- ✓ High cost of leasing fuel / maintenance etc.
- ✓ Little profit / low yields
- ✓ Small fields / farms, machinery cannot be used
- ✓ Lack of skills
- ✓ Cannot take loans

CHEMICAL FERTILIZERS

- ✓ These replace the nutrients that have been used up by the crop.
- ✓ With the use of fertilizers, nutrients are available for plant growth resulting in higher yields

Advantages

- ✓ Increases yield
- ✓ Extension of farm
- ✓ Grows faster because it provides nitrogen, potassium, and phosphate.
- ✓ Better than dung

Disadvantages

- ✓ Costly
- ✓ Knowledge and skills needed
- ✓ Pollution of ground water
- ✓ Eutrophication
- ✓ Not readily available

How Chemical fertilizers damage natural environment?

- ✓ Cause pollution of rivers/watercourses (runoff from farmland containing chemicals washes into them)
- ✓ Eutrophication in rivers (nitrates/phosphates cause algal blooms which increase CO_2 /reduce O_2) which destroy aquatic life
- ✓ Causes ecosystem to be unbalanced through loss of species
- ✓ Overuse of fertilisers damages soil

PESTICIDES AND INSECTICIDES

- ✓ These are sprayed to kill insects, which damage crop growth by eating leaves etc.

Advantages

- ✓ To kill insects, pests, and viruses
- ✓ Prevent loss
- ✓ Increases the output and income.

Disadvantages

- ✓ Costly
- ✓ Knowledge and skills needed
- ✓ Also kill useful insects
- ✓ Sever effects in natural food chain

LABOUR

- ✓ Labour is all the human effort that is put in from the time of plantation of crop to the time of harvest.

Advantages

- ✓ Maintain machinery
- ✓ Spread fertilizers
- ✓ Ploughing
- ✓ Sowing
- ✓ Threshing
- ✓ Harvesting

Disadvantages

- ✓ Can't work 24/7
- ✓ Less perfection in work if compared with machinery

SEEDS

- ✓ A desi variety of seeds produces lower yields as compared to a high yielding variety of seeds.
- ✓ But it must be noted that only desi varieties are adapted to the local climatic conditions.
- ✓ Therefore to achieve high yields in an adverse climate a hybrid of the two varieties is required

Advantages of Desi

- ✓ Cheap
- ✓ Less Chemical Fertilizer required if compared with HYV
- ✓ Locally available

Advantages of HYV

- ✓ Increases yields
- ✓ Grow's faster
- ✓ Less water demand
- ✓ Pest resistant
- ✓ Disease Resistant
- ✓ Healthy Growth
- ✓ Meet requirements of International standards
- ✓ Better Germination

Disadvantages of Desi

- ✓ Less yield
- ✓ Grow slowly if compared with HYV
- ✓ More water demand
- ✓ No Pest resistant
- ✓ No Stronger stems

Disadvantages of HYV

- ✓ To be imported
- ✓ Expensive to use
- ✓ Professional skills required
- ✓ Require more chemical fertilizers

- ✓ Seeds have to be bought every year/cannot sow seeds produced from crops grown

TRANSPORT

- ✓ To takes crops to factories
- ✓ Faster speed e.g. sugar cane to the mill to prevent loss of sugar content
- ✓ Dry ports for inputs (Fertilizer, HYV)

ELECTRICITY/DIESEL

- ✓ Tube wells
- ✓ Tractors
- ✓ Drying crops
- ✓ Other machineries e.g. Generator

HERBICIDES

- ✓ Kill weeds

IRRIGATION

- ✓ Better water supply
- ✓ Meet rain shortages
- ✓ Reduces salinity
- ✓ Increases yield

EDUCATION SKILL

- ✓ Machine repairing
- ✓ New seeds varieties
- ✓ Better methods
- ✓ Modern methods
- ✓ Avoid crop failure

To what extent is it possible to increase agricultural production by the use of modern methods? [6]

Possibilities

- ✓ More growth with fertilisers
- ✓ Less damage with pesticides
- ✓ More yield with better seed / HYVs crops
- ✓ HYVs are pest resistant
- ✓ Benefits of machines (Quicker, no need to rest, Perfect)
- ✓ Treatment of waterlogging and salinity e.g. with tubewells
- ✓ Training and education

Problems

- ✓ Lack of education means less training
- ✓ Lack of money to invest
- ✓ Traditional farming methods
- ✓ Over-use of irrigation water causes waterlogging / salinity
- ✓ High cost of fertiliser, machinery etc.
- ✓ Water pollution from runoff with fertiliser / pesticide
- ✓ May be unsustainable

PROCESSES

- ✓ A series of things that are done in order to achieve a particular result.

Ploughing

- ✓ A large piece of farming equipment with one or several blades pulled by a tractor or animal. It is used for digging and turning over soil specially before seeds are planted.

Sowing

- ✓ Spreading seeds in or on the ground with machine or by hands.

Irrigation

- ✓ To supply water to an area of land through tube well / channels so that crops will grow.

Fertilizing

- ✓ Adding substances to soil to make plants grow more successfully.

Weeding

- ✓ Cutting of wild plants growing where they are not wanted especially among crops.

Threshing

- ✓ To separate grains of wheat or rice from the chaff.

Harvesting

- ✓ Cutting the crops

WHEAT

Type

- ✓ Rabi crop

Uses

- ✓ Manufacture of bread and a variety of baked products
- ✓ Low grades of wheat are used as feed for livestock

By Products

- ✓ Chaff, used for feed and mix it with mud to make storage hut

Method Of Cultivation

- ✓ In oct-dec after ploughing the field, wheat seeds are sown directly into the ground
- ✓ Most of the farmers irrigate land twice.
- ✓ First irrigation, one month after sowing
- ✓ Second irrigation one month before harvesting
- ✓ Harvested after three months

Natural Inputs

- ✓ 10-20 degree for growing
- ✓ 25-30 degree for ripening
- ✓ Light rain in oct-nov for growth
- ✓ Requires 325-625 mm
- ✓ Alluvial, loam, clay, water retentive and well drained soils
- ✓ 90 days
- ✓ Little/No rain just before harvesting.

What grows in October-May. Explain why the climate at this time is most suitable for wheat farming [3]

- ✓ Mild temperatures / 10-20°C for growth
- ✓ Warmer / 25-30°C (in spring) for ripening / harvesting
- ✓ Moderate rainfall

- ✓ Dry period / no rain (in spring) for harvest

State two natural inputs necessary for wheat production, and for each explain its importance.

- ✓ Cool/moderate temperatures 10 - 20 - for germination and good growth
- ✓ Warmer; 25 - 30 for ripening
- ✓ Dry period - for ripening/harvesting
- ✓ Moderate rainfall for germination
- ✓ Alluvial soil - for good growth
- ✓ Flat land - for machinery and/or irrigation

Human Inputs

- ✓ Capital
- ✓ Machinery
- ✓ New HYV's / Desi Seeds
- ✓ Chemical fertilizers / Cow Dung
- ✓ Pesticides
- ✓ Irrigation
- ✓ Labour (Skilled and Unskilled Both)

Areas

- ✓ Nawabshah
- ✓ Nausharo
- ✓ Rahimyar khan
- ✓ Bhawalpur
- ✓ Multan

Hyv

- ✓ Maxi pak
- ✓ Shah khan 95
- ✓ Wadnak 95
- ✓ Kohsan 95

BARANI FARMING

- ✓ It is practiced in areas like the Potwar Plateau which have low amounts of seasonal rainfall.
- ✓ Important crops grown are wheat, maize, millet etc all which require low amounts of water and sunlight
- ✓ When the rains arrive or are about to arrive, the land is ploughed so it becomes soft.
- ✓ Immediately after the rain, the seeds are sown and the periodic cycle of sunny weather in between light rainy days continues till the harvest.
- ✓ Within the growing period hoeing is done (to remove weeds), if pesticides and fertilizers are available then they are added otherwise cow dung is used.
- ✓ Lastly the harvest season must be sunny and dry.

Describe the methods of cultivation of wheat on barani (rain-fed) lands.

- ✓ ploughing/preparation of soil is done when rain is expected
- ✓ when the rain falls in winter / October-November
- ✓ seed sown
- ✓ pesticides, fertilizers are used
- ✓ harvested when ripe / after 3-6 months

RICE

Type

- ✓ Kharif

Uses

- ✓ Used as food
- ✓ Exported

By Products

- ✓ Rice husks are used for making cardboards or covering roofs of houses after mixing it with mud
- ✓ For livestock

Method Of Cultivation

- ✓ The tractor pulls a plough in march. Each time he ploughs, he goes in a different direction, so that soil is well broken up.
- ✓ If the farmer cannot afford to hire tractor, he uses a bullocks to pull a plough at the right depth in the soil.
- ✓ While land is being prepared, rice seeds are sown in the nursery in the corner of the field.
- ✓ When seedlings are about 30 cm tall, they are ready to be transplanted.
- ✓ In may, farmers usually employ some labour for transplanting.
- ✓ Bundles of rice seedling are carried from nursery to prepared fields.
- ✓ Prepared fields have now been flooded to a depth of 25 cm, ready for transplanting the seedlings
- ✓ It is skilled work because seedling will float away if they are not planted properly.
- ✓ Rice crops are protected from birds
- ✓ Fertilizers is scattered on the field
- ✓ Insecticide is also scattered on the field
- ✓ In September, rice is ripe and ready for harvesting
- ✓ Water is drained off
- ✓ Sickel is used to cut the crop
- ✓ Rice is then tied in bundles and carried to be threshed manually.

Explain why the cultivation of rice is labour intensive.

- ✓ nursery for seeds
- ✓ repairing bunds for water
- ✓ prepare fields by ploughing
- ✓ flooding/irrigation
- ✓ transplanting seedlings
- ✓ fertiliser for nutrients/good growth
- ✓ pesticides to kill pests/for better growth
- ✓ drain water
- ✓ cutting/harvesting ripe crop

Natural Inputs

- ✓ Warm hot temperature 25 to 35 degree celcius with no cold season
- ✓ At least 1270 mm per year
- ✓ Over 2000 mm per year is ideal
- ✓ Dry sunny weather at harvesting time
- ✓ Loamey or clayey soil
- ✓ Flat Land

Explain how rainfall is linked to the processes?

- ✓ June Rain to soften soil for preparation of field/ploughing
- ✓ June-July Rain for planting seeds/seedlings
- ✓ June-September High/increasing rainfall for flooding fields
- ✓ June-September Sufficient rainfall/rain continues for growth
- ✓ September-October Drier period for harvest

Human Inputs

- ✓ Capital
- ✓ Machinery
- ✓ New HYV's / Desi Seeds
- ✓ Chemical fertilizers / Cow Dung
- ✓ Pesticides
- ✓ Irrigation
- ✓ Labour (Skilled and Unskilled Both)

Areas

- ✓ Larkana
- ✓ Sialkot
- ✓ Gujranwala

Name the provinces where most rice is grown.

- ✓ Punjab and Sindh

Hyv

- ✓ Basmati
- ✓ IrriPak

COTTON

Type

- ✓ Kharif

Uses

- ✓ Most widely used textile fibre in Pakistan.
- ✓ Clothes, fabrics

By Products

- ✓ Cotton seeds, separated from lint, a fluffy mass of fibres inside cotton balls
- ✓ Cotton seeds are used as animal feed and for the extraction of oil.

Method Of Cultivation

- ✓ Sown at a distance apart of 30 cm to 45 cm in April-May.
- ✓ One month later fields are irrigated.
- ✓ Second irrigation takes place after a further two months
- ✓ Cotton ball ripen in dry months of Oct and Nov.
- ✓ Plant reaches a height of up to 135 cm-150 cm.
- ✓ After picking cotton balls are loaded on to trucks.

Natural Inputs

- ✓ Ideal temperature for cultivation is 25 to 35 degrees.
- ✓ Mid nights temperature are better for development of the balls.
- ✓ Sensitive to frost.
- ✓ 1000 mm rainfall.
- ✓ Medium loam.
- ✓ Level land
- ✓ Little rain for ripening/harvesting

Human Inputs

- ✓ Capital
- ✓ Machinery
- ✓ New HYV's / Desi Seeds

- ✓ Chemical fertilizers / Cow Dung
- ✓ Pesticides
- ✓ Irrigation
- ✓ Labour (Skilled and Unskilled Both)

Areas

- ✓ Nawabshah
- ✓ Bhawalpur
- ✓ Multan

Hyv

- ✓ Nayyab 78
- ✓ B-557
- ✓ 149-F

What Harms The Cotton Crop?

- ✓ Greatly affected by rise in day and night temperature changes to leaf-curl and fruit shedding.

Why is cotton not grown further north?

- ✓ Too cold (in summer / growing period)
- ✓ Sensitive to frost
- ✓ Rain / too wet during harvest
- ✓ Poor soil / infertile etc.
- ✓ Steep slopes / no flat land
- ✓ Remote / long way from factories

Why is cotton not grown further west?

- ✓ Too dry / lack of rainfall (for growth)
- ✓ Lack of irrigation canals
- ✓ Too cold (in growing period)
- ✓ Poor soil / infertile / etc.
- ✓ Steep slopes / no flat land
- ✓ Remote / long way from factories

SUGAR CANE

Type

- ✓ Kharif.

Uses

- ✓ Sugar.
- ✓ Brown sugar.
- ✓ Gur

By Products

- ✓ Bagasse, fibre left over. It is used to make paper, clipboard and animal feed.
- ✓ Molasses, liquid left over after crystallization of sugar. It is raw material for citric acid, yeast, rubber. It is also used as a fuel to generate electricity in sugar-mills.

Methods Of Cultivation

- ✓ Sugar-stalks 30 cm high are planted in April-May.
- ✓ A distance of 30 cm is kept between each stalks.
- ✓ Quality depends upon frequency of irrigation and fertilizers.
- ✓ If land is well irrigated then plant rise upto height of 6.7 feet and the crop can be ratooned and so harvested for 2-3 successive years.

Natural Inputs

- ✓ Temperature 25-35 degree celsius
- ✓ Hot growing season.
- ✓ Dry harvesting period.
- ✓ Atleast 500 mm rainfall.
- ✓ Over 1500 mm is ideal.
- ✓ Rain should not be excessive during ripening period because it may dilute sugar content.
- ✓ Alluvial soil.
- ✓ Deep and Firm soil to support tall stem.

Human Inputs

- ✓ Capital
- ✓ Machinery
- ✓ New HYV's / Desi Seeds
- ✓ Chemical fertilizers / Cow Dung
- ✓ Pesticides
- ✓ Irrigation
- ✓ Labour (Skilled and Unskilled Both)

Areas

- ✓ Nawabshah.
- ✓ Faisalabad.
- ✓ East central Punjab.

Give two reasons why sugar cane factories should be built as close as possible to the fields where sugar cane is grown.

- ✓ Loses its sugar content after harvesting
- ✓ Heavy/bulky to transport
- ✓ Saves transport cost

Explain how this crop is processed.

- ✓ taken to factory/mill
- ✓ quickly/without delay
- ✓ washed/scrubbed
- ✓ crushed
- ✓ juice collected
- ✓ refined
- ✓ crystallised
- ✓ whitened/made into white sugar
- ✓ baggase produced (a waste product)

Hyv

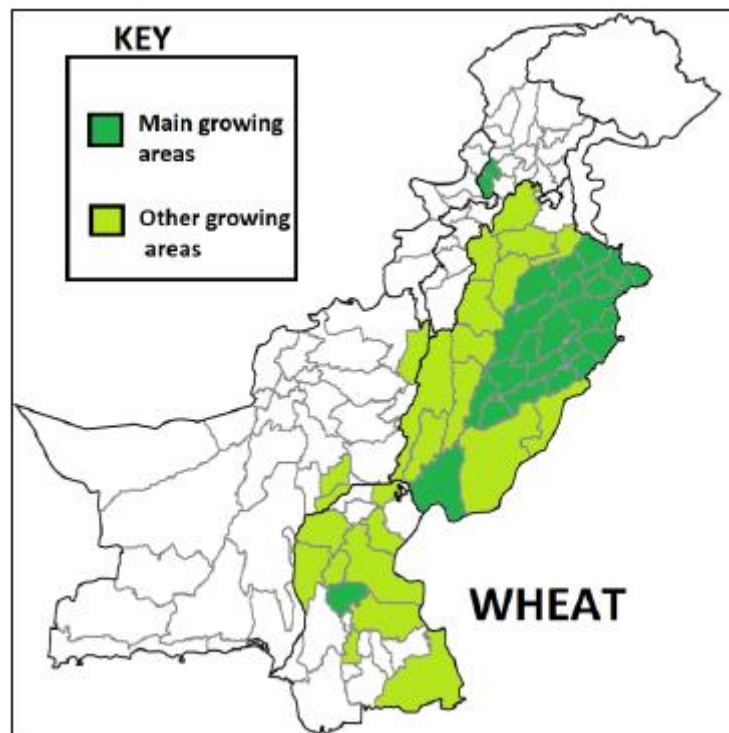
- ✓ JN-88
- ✓ Thatta-10

Explain why the production of agricultural crops varies from year to year.

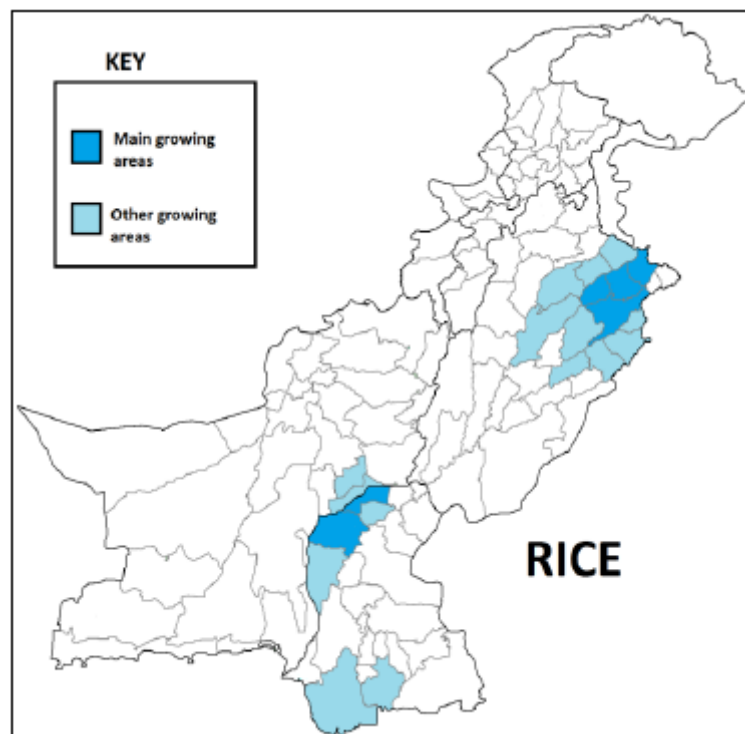
- ✓ temperatures vary
- ✓ rainfall varies,
- ✓ e.g. floods, drought, extreme events
- ✓ irrigation water may be short
- ✓ pests/disease/virus
- ✓ quality of inputs depends on last year's profit
- ✓ human factors, e.g. sickness
- ✓ changes in government policies

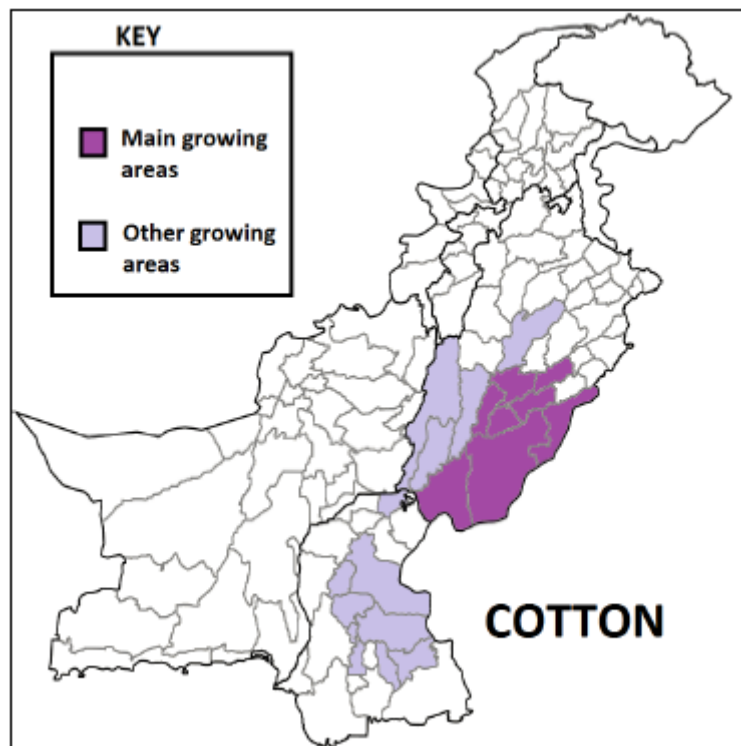
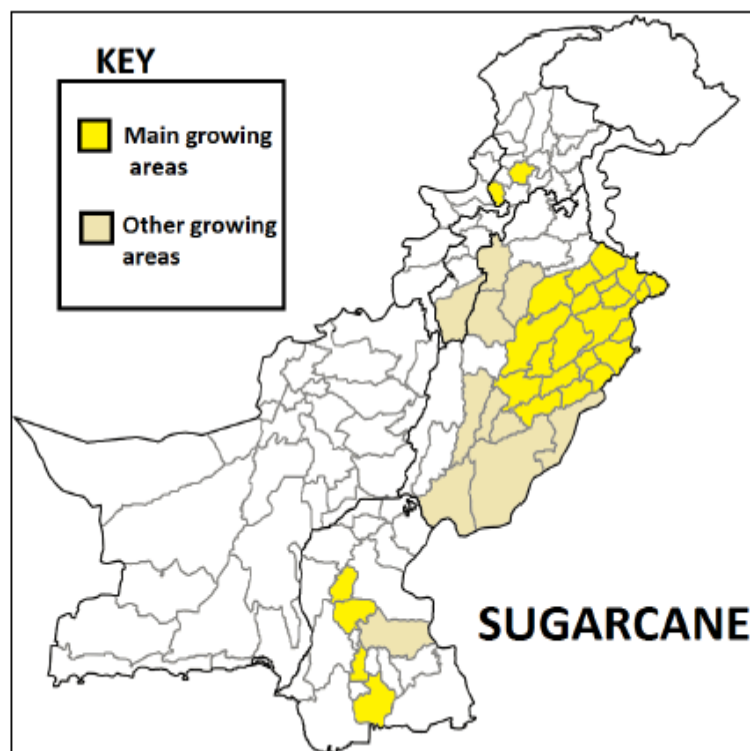
MAPS OF WHEAT/RICE/COTTON/SUGARCANE

Wheat



Rice



Cotton**Sugarcane**

WHEAT/RICE/COTTON/SUGARCANE

Wheat



Rice



Cotton



Sugarcane



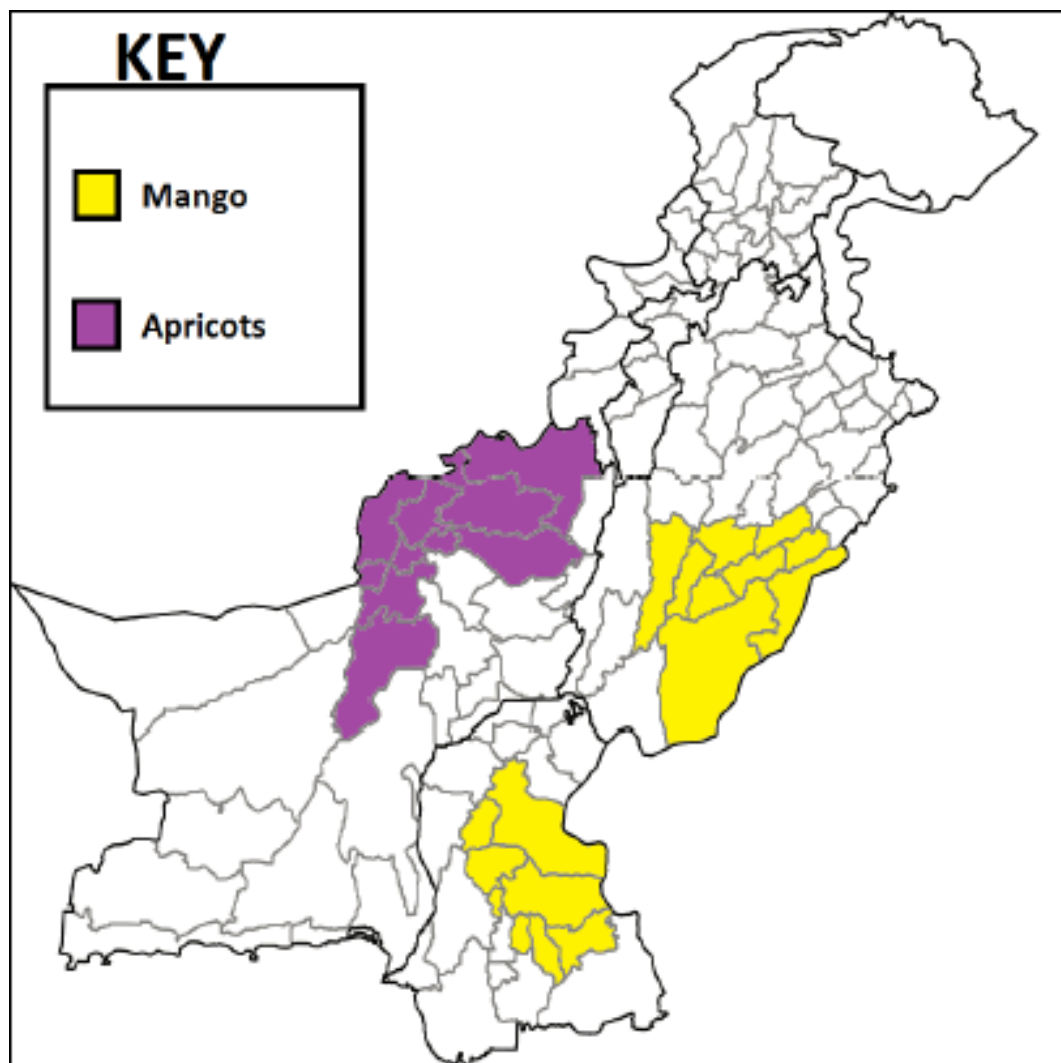
APRICOTS

- ✓ Apricots are grown mainly in areas of mid-northern Balochistan.
- ✓ Pakistan is the 4th largest producer of Apricots.
- ✓ The apricots need to fulfil chilling units, meaning that they must be exposed to cold for a certain period of time.
- ✓ Winters must be cold (but not colder than -30°C) to allow for proper dormancy.
- ✓ Dry weather must exist at the time of maturity/harvest.
- ✓ It is important that there should be no sudden and dramatic change in temperatures in spring which kills the flowers.
- ✓ Apricots grown in well drained soils with PH around 6.5



MANGOES

- ✓ Mango requires a wet hot summer season with rainfall around 250mm.
- ✓ This needs to be followed by a long dry winter.
- ✓ Dry winter means that the plant is less susceptible to attacks by fungus etc.
- ✓ It needs deep well drained loamy soil.
- ✓ Mangoes are grown in southeastern Punjab and eastern Sindh.



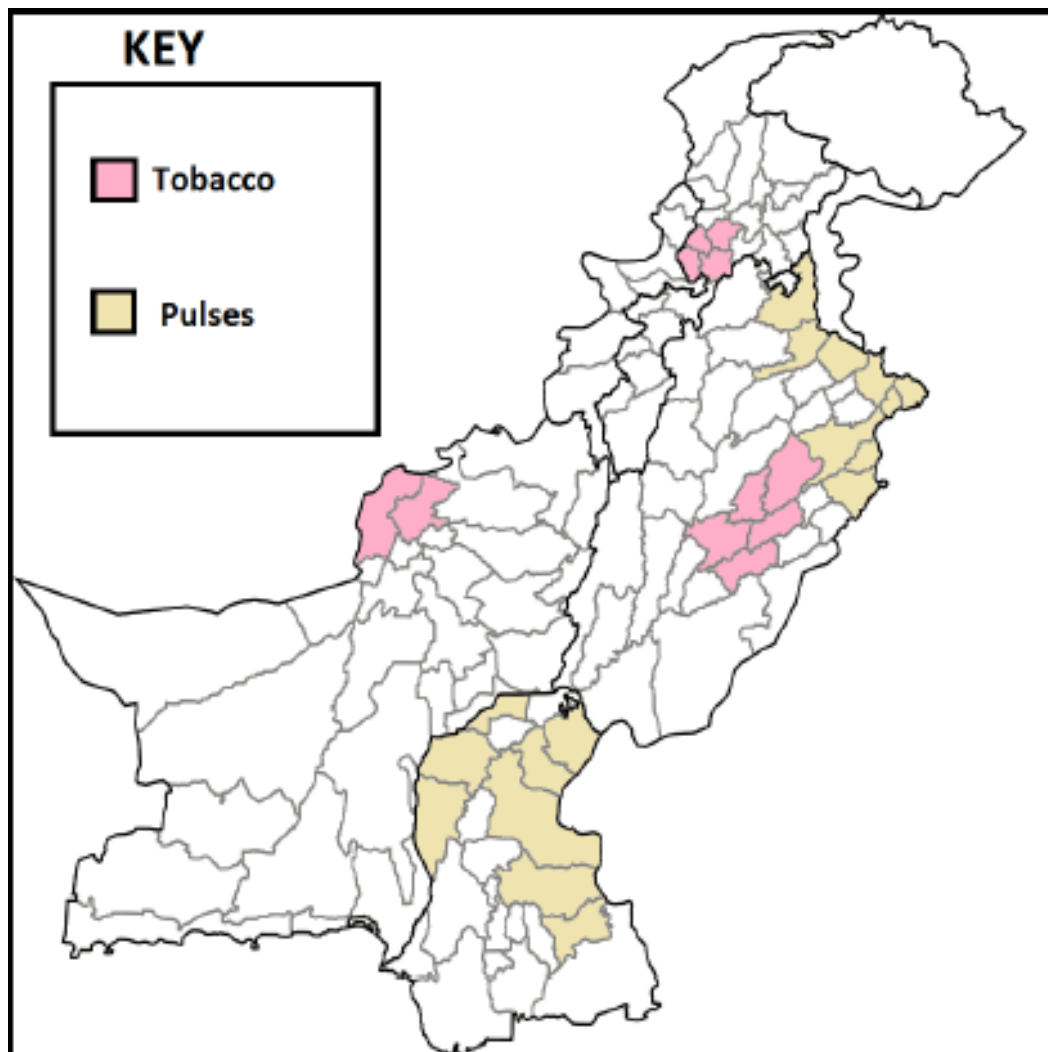
TOBACCO

- ✓ Tobacco plants are usually first grown in nurseries etc and then transplanted in the fields when the risk of frost has passed.
- ✓ They need light rainfall early on, followed by a period of bright sunshine with rains in between, followed by a dry period at harvest
- ✓ Tobacco is grown in central parts of Punjab, northern Balochistan and central parts of Khyber-Pakhtunkhwa



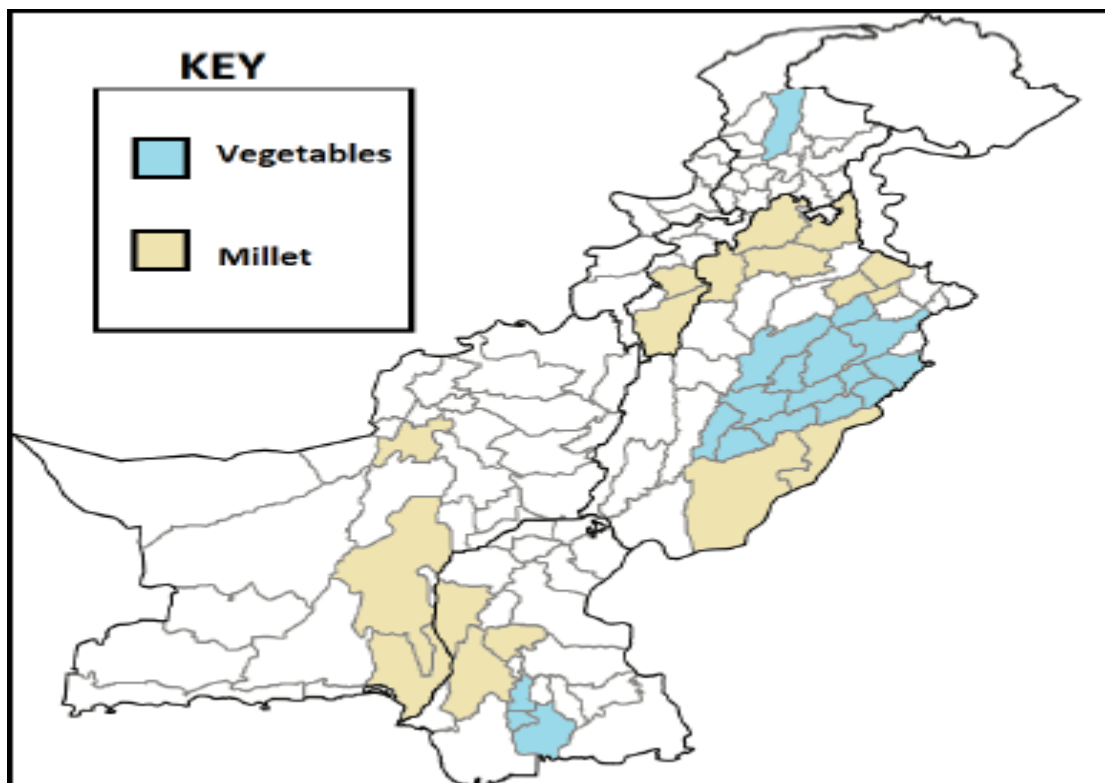
PULSES

- ✓ Pulses require high temperatures but are highly susceptible to frost.
- ✓ They can also tolerate high rainfall provided it doesn't come at time of pollination etc.
- ✓ Also the soil must not be waterlogged or saline. Pulses can be grown on sandy, loamy etc soils
- ✓ Pulses are grown in eastern and western Sindh, along with north eastern Punjab



MILLET

- ✓ Millet requires moderate rainfall and is sensitive to frost.
- ✓ It will not tolerate waterlogged soils or harsh drought.
- ✓ They are grown in southeastern and northwestern Punjab, western Sindh, and central and southeastern Balochistan



DATES

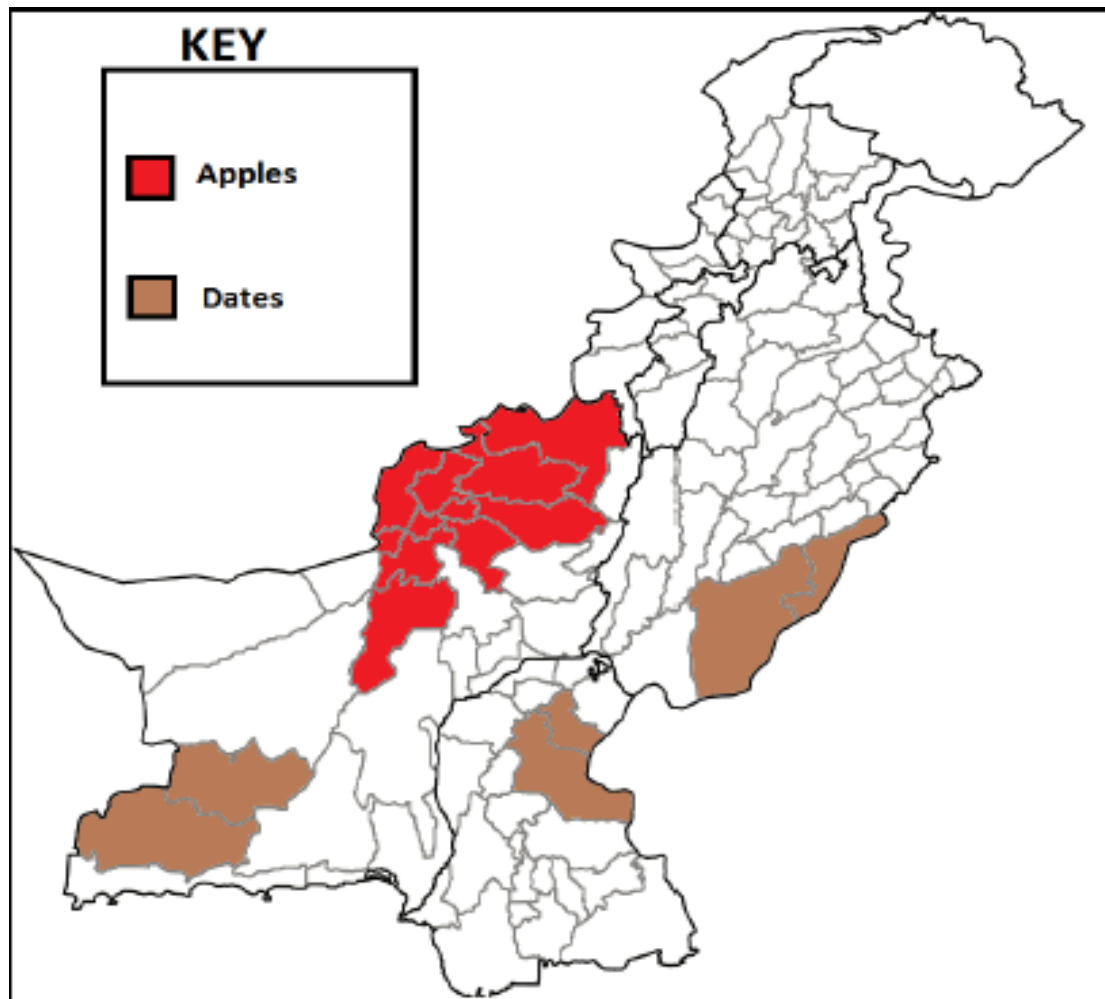
- ✓ Dates are grown in parts of Balochistan and in parts of Tharparkar desert (southern Punjab and eastern Sindh).
- ✓ It needs long hot summers with high day and night temperatures.
- ✓ It can tolerate fluctuations in temperature whether cold or hot.
- ✓ Mild winters and a dry sunny time for harvest is also required.
- ✓ They can grow in salty soils but they must be well drained



APPLES

- ✓ Apples are solely grown in northern Balochistan.
- ✓ They have the highest requirement of chilling units in fruits.
- ✓ Winters must be cold to allow for proper dormancy.
- ✓ These must be followed by rains during the growing season.
- ✓ Dry weather must exist at the time of harvest.
- ✓ Well drained loamy soils are best for growing apples





CITRUS FRUITS

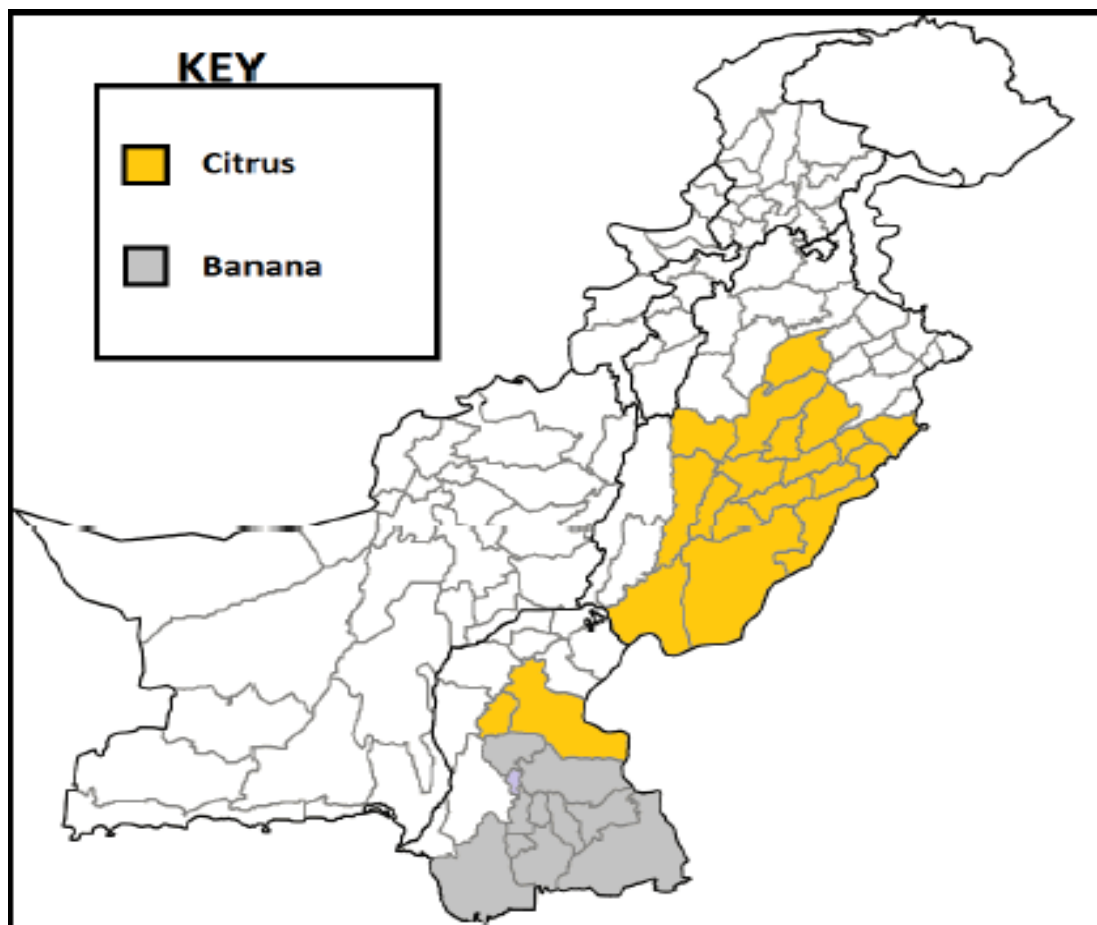
- ✓ They grow in tropical or subtropical climate with hot summers and moderate rainfall.
- ✓ They are also sensitive to frost and strong winds.
- ✓ Most of the production is from Punjab including its central but mostly southern parts.
- ✓ Some citrus production also occurs in eastern parts of Sindh



BANANAS

- ✓ Bananas are grown exclusively in southern Sindh.
- ✓ They require a hot dry season lasting for around 2-3 months, with a mean rainfall of around 10 cm.

- ✓ Bananas are very sensitive to frosts, which can suspend maturity or even kill the plant.
- ✓ They are also very vulnerable to strong winds, which damage the fruit etc.
- ✓ They require well drained alluvial soils

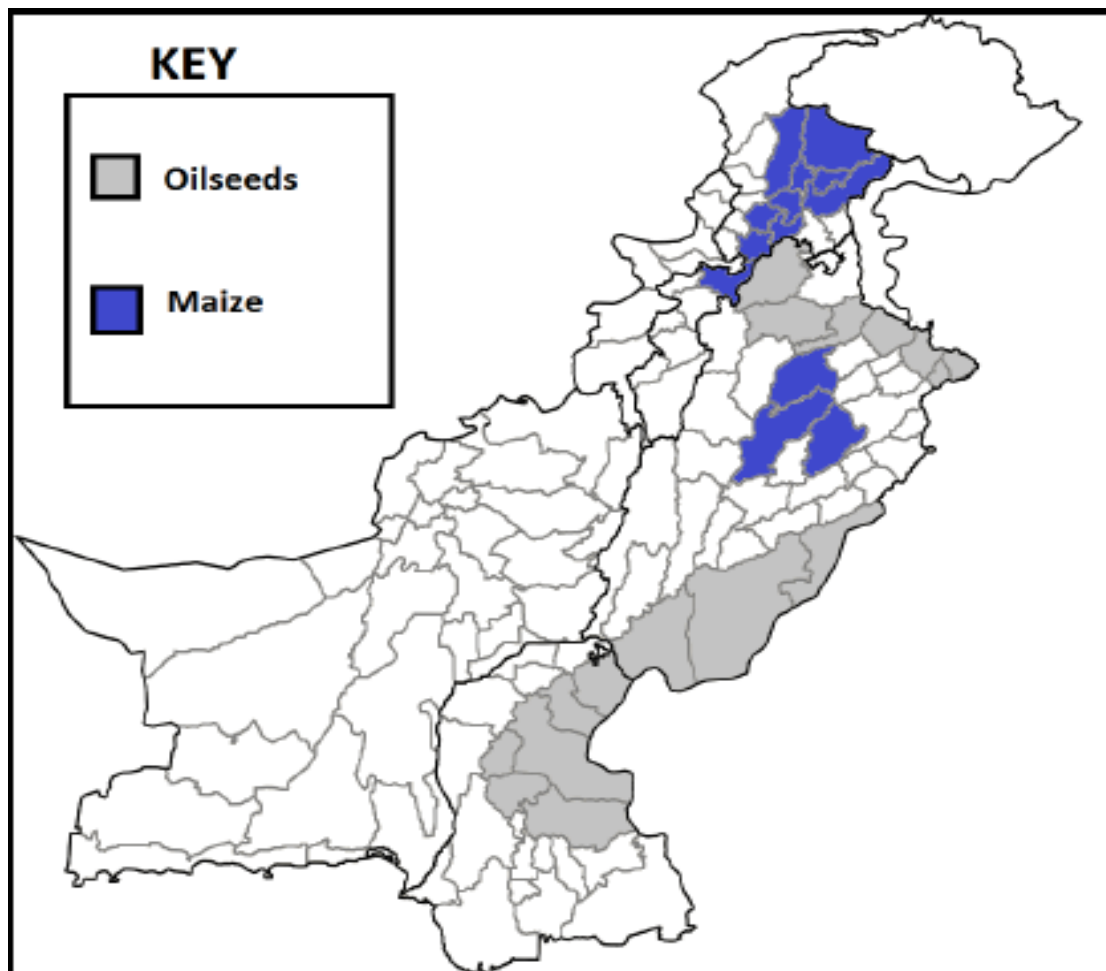


OILSEEDS

- ✓ Oilseeds usually require average temperatures varying from 20-30°C.
- ✓ It must also be noted that high temperatures can hamper or delay growth and that frost kills the plants.
- ✓ Oilseeds are tolerant to drought for some periods and require well drained deep alluvial soils
- ✓ They are grown in southern parts of Punjab and eastern parts of Sindh, along with some northern parts of Punjab.

MAIZE

- ✓ Maize needs a hot bright growing season to flourish.
- ✓ It is very intolerant to frost and needs moderate rainfall well distributed throughout growth.
- ✓ It also needs well drained deep alluvial soils
- ✓ It is grown in central Punjab and central parts of Khyber-Pakhtunkhwa



Name two crops that are used mainly for animal feed.

- ✓ maize, millet, oil seeds

Name one crop that is not a food crop.

- ✓ Tobacco

Name one crop that is rich in protein.

- ✓ Pulses

Name one type of oil seed.

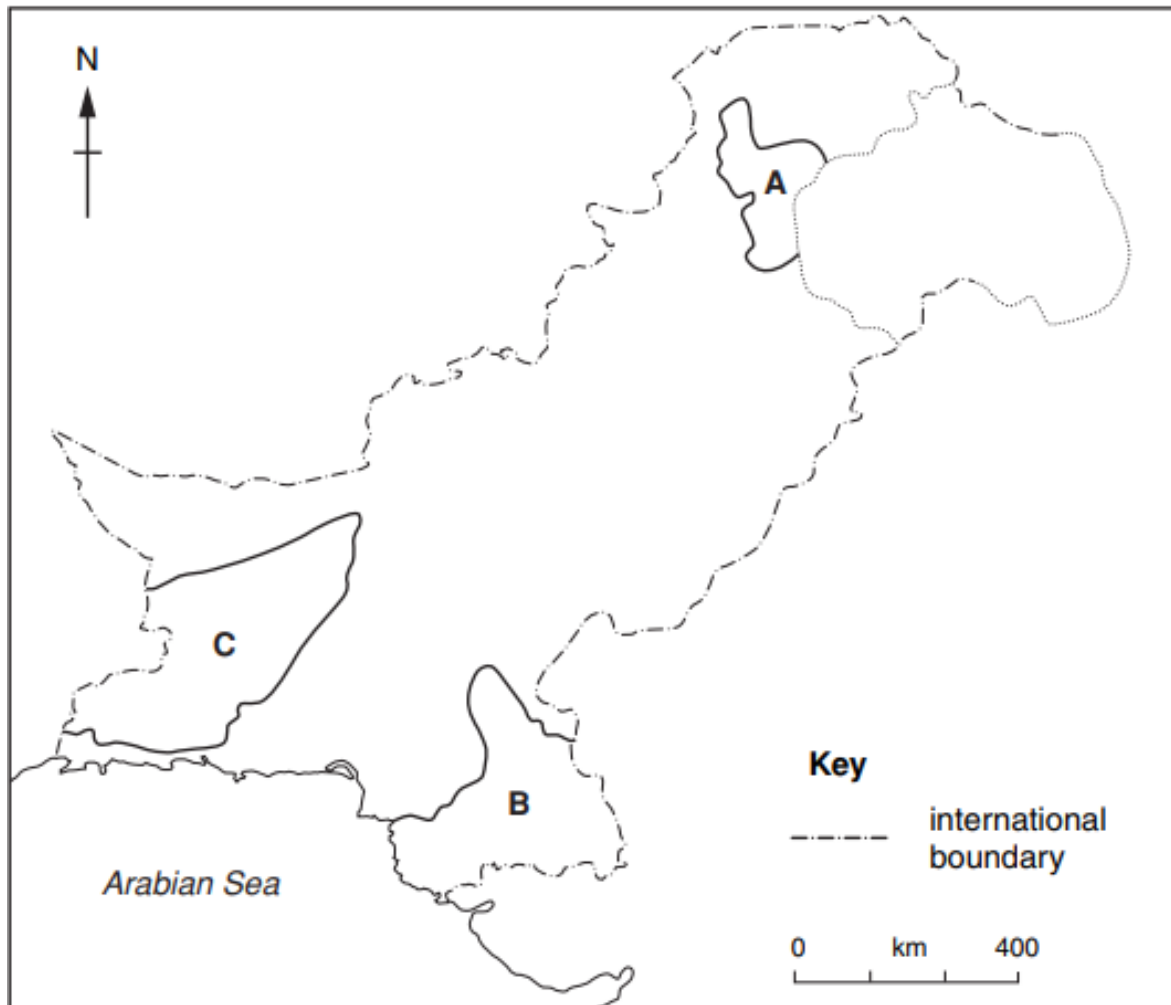
- ✓ Rape, mustard, groundnut, sesame, sunflower, safflower, soya bean

Why are fruit crops grown in mountain valleys?

- ✓ warmth
- ✓ shelter
- ✓ sunshine
- ✓ rain/less snow
- ✓ soil
- ✓ flat land

Why are fruit crops grown mainly for local use?

- ✓ Perishable
- ✓ Heavy to transport
- ✓ Small amounts
- ✓ Not of export quality



Name the two main fruit crops grown in area A.

✓ apples, apricots, almonds

Name one of the main fruit crops grown in area B.

✓ Bananas/mangoes/citrus fruit

LAND REFORMS

- ✓ The principle of dividing land for farming into small pieces so that more people can own some.

What did the land reform laws aim to do?

- ✓ Redistribute land more equally
- ✓ Take land away from large landowners/landlords and give it to the tenants/poor farmers

LAND CONSOLIDATION

- ✓ Land consolidation is a planned readjustment and rearrangement of fragmented land parcels and their ownership.
- ✓ It is usually applied to form larger and more rational land holdings.
- ✓ The government has introduced land reforms to consolidate the holdings of farmers, so that their far flung fields can be combined to form one big economically viable field.
- ✓ It thus becomes easier to use machines on it, obtaining loans becomes easier (as banks are more confident that farmer will be successful after using their money and will be able to repay the loan), farmer can experiment on his field and can ultimately save the time and hassle of working on multiple farms (this allows him more time to improve things like water supply to a single farm etc, all of which increase crop yields even further)

LAND FRAGMENTATION

- ✓ Land fragmentation can be defined as a situation where a farming household possesses several non-contiguous land plots, often scattered over a wide area.

GOVERNMENT EFFORTS

- ✓ Offering loans (for machinery / tubewells / fertilisers / pesticides / seeds / labour costs)
- ✓ Providing large irrigational schemes such as dams and canals.
- ✓ Providing solutions for waterlogging such as SCARP
- ✓ Increasing the production of fertilizer.
- ✓ Distribution of improved seeds and regulation of quality seeds through government departments.
- ✓ Developing plant protection programmes including regular checks for detection of pests.
- ✓ Providing advisory services.
- ✓ Educating / training farmers (on use of HYVs / modern farming methods / sustainable methods / organic farming)
- ✓ Subsidies eg. for imported machinery, fertiliser prices lower

Why these actions not possible?

- ✓ Lack of money
- ✓ Illiteracy
- ✓ High population
- ✓ Other calls on government investment / attention
- ✓ Fears of unemployment due to mechanisation
- ✓ Land reforms may fail due to corruption / power of landlords etc.

SUSTAINABLE AGRICULTURE

- ✓ Supplying the agricultural product needs of the present generation while protecting agricultural product needs of those in the future

Possibilities

- ✓ Less overcropping / multicropping
- ✓ Methods of preserving soil e.g. terraces / contour ploughing
- ✓ Soil management through afforestation projects.
- ✓ Restrict use of heavy machinery
- ✓ Keeping vegetation cover
- ✓ Better water management / avoiding over watering / conserving water / lining canals
- ✓ Organic farming / using manure
- ✓ Use of appropriate knowledge / training
- ✓ Crop rotation.
- ✓ On farm waste recycling.
- ✓ Weed control by marching.
- ✓ Pest and disease control.
- ✓ Reclamation of deserts with help of irrigation schemes.

Difficulties

- ✓ High demand for more food
- ✓ Pressures on land e.g. for timber
- ✓ Lack of education and less awareness of sustainable methods
- ✓ Unco-operative landlords
- ✓ Land reform needed
- ✓ Lack of government will / support / investment
- ✓ Resistance to changing traditional / modern methods

LIVESTOCK FARMING

- ✓ This type of farming (also known as pastoral farming) is concerned with rearing of animals, whose products are then sold.

IMPORTANCE OF LIVESTOCK FARMING

- ✓ Draft power(e.g. Persian Wheel)
- ✓ Transport in rural areas
- ✓ Food /meat/milk/eggs (for fast growing population)
- ✓ Source of Employment, Income
- ✓ Dung as manure
- ✓ Dung as a fuel source (domestic or biogas)
- ✓ Source of raw materials (hides/skins/wool/hair/bones especially for cottage industries/export potential/food processing industries)

LIVESTOCK CAN INCREASE FOOD SUPPLY

Possibilities

- ✓ Provides meat / milk / eggs / cheese / named food item
- ✓ Provides protein
- ✓ Provides raw material for food processing industry
- ✓ Through selective breeding or better husbandry we can have more animals which will increase food supply.
- ✓ Through fattening programmes we can have more meat and milk.

Not Possible

- ✓ Lack of land for Livestocks as too much land is used for agriculture
- ✓ Lack of education/Skills
- ✓ Cost of modern methods / facilities
- ✓ Lack of government support

DISADVANTAGES OF LIVESTOCK FARMING

- ✓ Loss of land / water for food crops.
- ✓ Overgrazing problems.

- ✓ Less investment in other forms of farming.
- ✓ Disposal of waste / problems of cleanliness /
- ✓ Disease transfer to humans
- ✓ Some products not of export quality / banned by western countries

NOMADIC FARMING

- ✓ Nomads are the people who move from one place to another place along with their livestock in search of water and pasture.
- ✓ Once they find these resources and settle over there and utilize their resources
- ✓ They move to another place so that's why they keep on moving for the search of new pastures.
- ✓ They hardly return to the old pastures unless rainfall takes place for new pastures to grow.
- ✓ The area in which they graze have very poor pastures because of extreme arid conditions.
- ✓ They keep sheep, camels and goats because they can survive in arid conditions and can survive in poor pastures as they can nibble the thin grass.

Advantages

- ✓ Free access to pastures from open fields.
- ✓ Low cost of inputs.
- ✓ Source of income.
- ✓ Dung for soil.
- ✓ Don't need to hire labour.

Disadvantages

- ✓ Don't have veterinary facilities.
- ✓ Don't have permanent places to live.
- ✓ Have to search for food for long distance.
- ✓ Unreliable income.
- ✓ Soil erosion.

Inputs

- ✓ Livestocks.
- ✓ Pastures from open fields.
- ✓ Water from oasis, wells, karez, ponds, lakes.
- ✓ Tents for shelter.
- ✓ Family labour.

Processes

- ✓ Natural breeding.
- ✓ Grazing.
- ✓ Migration for water, pasture.
- ✓ Milking manually.
- ✓ Collecting manure for fertilizer, fuel.
- ✓ Preserving meat.

Outputs

- ✓ Meat.
- ✓ Wool.
- ✓ Livestock.
- ✓ Manure.
- ✓ Milk.
- ✓ Income.

Environmental Problems

- ✓ Soft erosion.
- ✓ Desertification.
- ✓ Deforestation (damage to young trees).

Why Many Nomads In Baluchistan?

- ✓ Shortage of resources.
- ✓ Cultivation of crops is difficult or impossible.
- ✓ Rugged landscape.
- ✓ Population density is least so plenty of land is available.

TRANSHUMANCE

- ✓ It is seasonal migration along with the livestock especially in winters from highland areas to the valley.
- ✓ When the summer approaches they move back to the high land pastures, but below snow line.
- ✓ In winter high land pastures are covered with snow and water sources also freezes so that's why they move to valley where temperature are suitable for them to survive and water, pastures are available.
- ✓ Sometimes they move to village market for selling their surplus such as animals, wools, skin and milk.
- ✓ In summer they move back to high land pastures below snowline because in summer rich pastures grow as snow and glaciers melt so therefore water is available.
- ✓ They are mainly found in N.mountain and western highlands
- ✓ From Himalayas they migrate towards to Kashmir valley.
- ✓ From Hindukush to Chitral and Swat valley.
- ✓ They keep sheep, goat, cattle, yak as these animals can adapt in highland climate and mountainous topography.

Advantages

- ✓ Good access to rich pastures.
- ✓ Good accessibility of water.
- ✓ Do not have to travel long distances.
- ✓ Source of income.
- ✓ No need to hire labour.
- ✓ Requirement of food and clothes fulfilled.

Diadvantages

- ✓ Lack of facilities.
- ✓ No permanent home.
- ✓ Unreliable income.
- ✓ Poor quality animals.

Inputs

- ✓ Livestock.
- ✓ Pastures.
- ✓ Water.
- ✓ Labour.

Processes

- ✓ Breeding.
- ✓ Grazing.
- ✓ Milking.
- ✓ Slaughtering.

Outputs

- ✓ Meat.
- ✓ Wool.
- ✓ Milk.
- ✓ Manure.

SETTLED

- ✓ They do not move, they have permanent homes.
- ✓ They are found in villages of sindhs and Punjab along rivers and on doabs.
- ✓ They keep cow, hens, goat, buffaloes, bullocks.

Advantages

- ✓ Own grazing ground known as shanilat.
- ✓ Do not move long distance.
- ✓ Reliable source of water.
- ✓ Proper sheds.
- ✓ All sources of inputs available.

Disdvantages

- ✓ Area which can be used for cultivating crop is utilized for grazing ground.
- ✓ Lack of nutritional fodder.
- ✓ Lack of investment.
- ✓ Poor quality animal.
- ✓ No experience.
- ✓ Unreliable source of income.

Inputs

- ✓ Grazing grounds.
- ✓ Water.
- ✓ Labour.
- ✓ Shed area

Processes

- ✓ Natural breeding.
- ✓ Milking manually.
- ✓ Collecting, manure, eggs.
- ✓ Slaughtering

Outputs

- ✓ Meat.

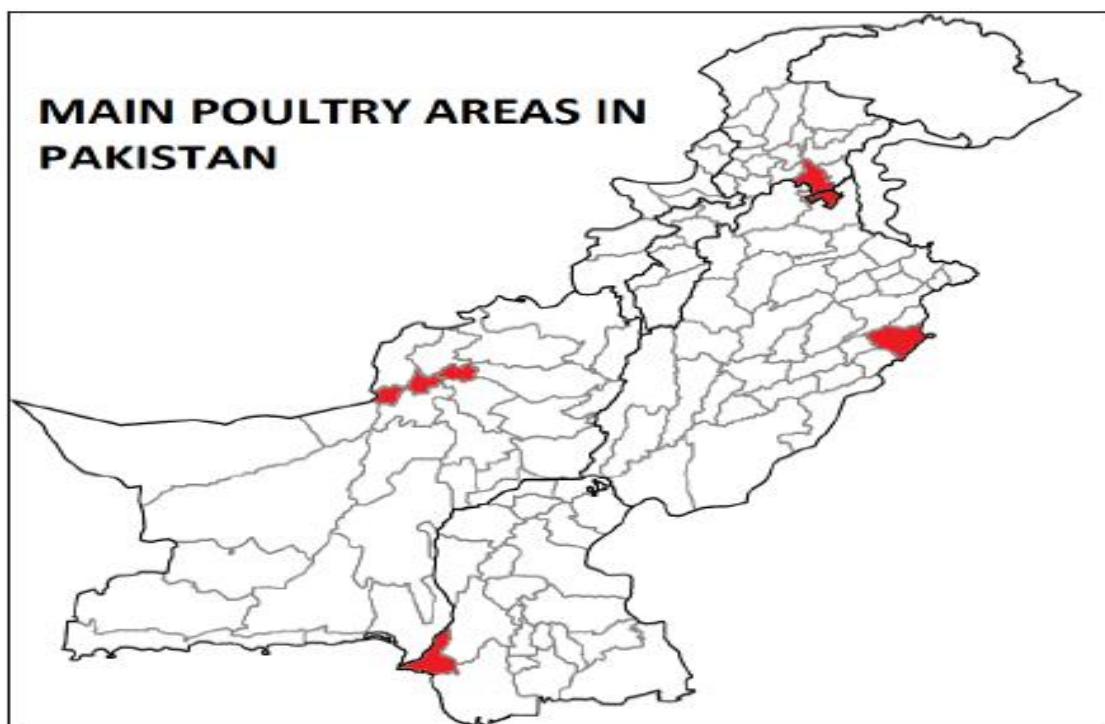
- ✓ Livestock
- ✓ Manure.
- ✓ Eggs.
- ✓ Milk.
- ✓ Wool.

POULTRY FARMING

- ✓ Poultry farming is the practice of raising poultry, such as chickens, turkeys, ducks, and geese, as a subcategory of animal husbandry, for the purpose of farming meat or eggs for food.
- ✓ In Pakistan, most of poultry farming consists of chicken.
- ✓ Poultry farms are mostly found around dense centres of population (Karachi, Quetta, and Lahore) and cooler areas (Murree, Abbottabad).
- ✓ Nearness to population centres reduces cost of transportation and cooler areas are preferred for optimal growth of chickens.

Steps Followed In Poultry Farming

- ✓ First the poultry farm building is cleaned properly, disinfectant is applied and the building is fumigated.
- ✓ Maize is used as food, when the chicks arrive they are given antibiotics in their food for 3-7 days.
- ✓ The temperature of the building is maintained between 32 and 37 degree Celsius.
- ✓ The chickens are put in cages so their eggs and wastes can be easily removed etc.
- ✓ After around 4-8 weeks the chickens can be slaughtered for their meat



BUFFALOES

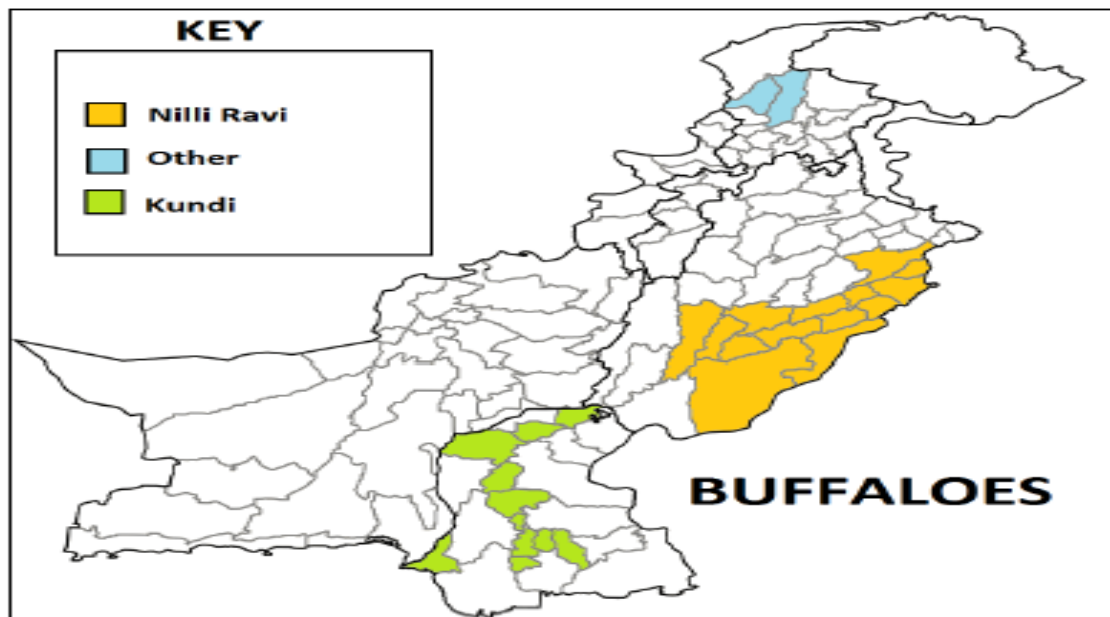
- ✓ Buffaloes are found mostly in canal fed areas of Pakistan, especially the doab between Ravi and Sutlej.
- ✓ Nilli-Ravi breed is found in Punjab, which is known for its high milk production.
- ✓ In Sindh, Kundi breed is found along both banks of Indus but mostly in Northern and central Sindh.
- ✓ Other breeds are found in areas of Khyber-Pakhtunkhwa
- ✓ Presence of water is very important as this animal needs to cool itself by smearing mud on its body.
- ✓ Water is also used for drinking and for cleaning the farm etc.
- ✓ A lot of fodder is also required (which becomes expensive to buy in non-agricultural areas) and thus buffaloes are not found in Balochistan as this would be too uneconomical.
- ✓ There are less urban areas there and thus less demand for beef.
- ✓ Buffaloes are considered to be black gold of Pakistan because the milk they produce has a higher fat content than that of cows and goats etc.
- ✓ At present, buffaloes provide almost 70% of the milk produced in Pakistan.
- ✓ Also, their meat is white and desirable due to the low cholesterol level as compared to cow's meat
- ✓ However, it must be noted that still there is ample room for improvement.
- ✓ Pakistan is the 2nd largest buffalo milk producer in the world after India.
- ✓ Also, Pakistan has one of the best breeds for producing milk (like Nilli and Ravi) but due to mismanagement, inefficient marketing system, old livestock technique, exposure to heat, shortage of fodder and late age of maturity mean that this potential is yet to be exploited

Explain why buffalo are not reared in Balochistan.

- ✓ Lack of water to drink
- ✓ Lack of water to wash / lie in / bath in / keep cool
- ✓ Lack of fodder crops. Buffalo need more food.
- ✓ Lack of demand / few urban areas

Suggest why buffalo farms can often be found around urban areas.

- ✓ Food (for urban population) / demand for milk or meat
- ✓ Must be fresh / deteriorates quickly
- ✓ Can make deliveries easily / supplied on a regular basis
- ✓ Products for processing, e.g. milk, ghee, butter



CATTLES

- ✓ Important cattle breeds are Red Sindhi and Sahiwal, which are internationally recognized for their milk production.
- ✓ Another important Breeds are bhagnari (Imp for draft power found in Punjab and sindh), Dhani (Imp for draft power found in N.Areas)
- ✓ Cattle are spread in Northern, central and Southern Punjab.
- ✓ In Sindh mainly in areas of Tharparkar desert.
- ✓ In Balochistan they are found in district of Hab and in Northcentral parts of Khyber-Pakhtunkhwa
- ✓ The yield of dairy animals in Pakistan is around 1/5 to 1/7 as to what yields are achieved in Europe and United States of America.
- ✓ If our yields can be improved this can save us from import of milk and related products, which costs around 20 million dollars annually.



GOATS

- ✓ Goats have a much wider distribution than sheep in Pakistan.
- ✓ They are found in almost whole of Punjab, Eastern and Southern Sindh, Makran coast and central Balochistan and districts of Peshawar and Mardan in Khyber-Pakhtunkhwa.
- ✓ Goats are also very adaptable like sheep but since their meat and milk is preferred over sheep; to satisfy this demand they are reared in larger numbers as compared to sheep

What environmental problems can be caused by keeping this type of livestock? Or Why government discourages rearing of Goats?

- ✓ Overgrazing
- ✓ Damage to young trees / deforestation
- ✓ Soil erosion / desertification / land becoming barren

SHEEP

- ✓ Sheep are mostly found in rugged areas of Northern Balochistan, Gilgit-Baltistan, Khyber-Pakhtunkhwa and parts of Southern Punjab.
- ✓ Sheep can survive both hot and cold seasons, and feed on shrubs and grasses.
- ✓ These adaptabilities make them vital for people living in these areas for meat, milk, wool, bones etc





SUBSISTENCE LIVESTOCK FARMING

- ✓ In it animals are kept to do daily tasks and provide daily food requirements usually by a subsistence farmer (agriculture one).
- ✓ Animals include chicken, buffalo, cow, sheep and goats albeit on a very small scale.
- ✓ Chicken provide eggs and meat, cows provide milk whereas the bull (male cow) is used for ploughing and transport (during planting, growth and harvest of crop).
- ✓ The manure is also used as fertilizer for crops.
- ✓ A pair of two bulls is known as a bullock.

COMMERCIAL LIVESTOCK FARMING

- ✓ Cows and buffaloes are raised in an organized manner for commercial purposes.
- ✓ It can be very profitable for small land owners, giving higher returns as compared to traditional farming methods, as land requirements are low.
- ✓ The fodder can be grown on the fields and then fed to dairy animals.
- ✓ The animals are also given a ration of special protein rich diet along with vaccinations to protect from infections etc, which can affect production of milk.
- ✓ If bigger dairy farms are established then the milk can be processed and packed too thus adding value to it.
- ✓ When the cow stops producing milk it is slaughtered along with other bulls for their meat, which is then sold in the market to earn a profit
- ✓ Dairy farms are important in Pakistan around big cities like Karachi etc.
- ✓ These farms provide milk regularly and cheaply due to the small distances involved.
- ✓ Meat and hides are also provided.
- ✓ Hides are used by leather industry and milk is used to make ghee.
- ✓ Furthermore, the dung can be sold and used in place of firewood etc

PROBLEMS OF LIVESTOCK

- ✓ Few veterinary facilities
- ✓ Lack of grazing grounds
- ✓ Expensive and difficult to keep in urban areas

- ✓ Lack of marketing facilities
- ✓ Lack of facilities for storage
- ✓ Old methods of breeding
- ✓ Unhygienic conditions
- ✓ Desertification due to overgrazing
- ✓ Lack of investment
- ✓ Disease transfer to humans.

HOW CAN LIVESTOCK BE IMPROVED?

- ✓ Capital, investment, loan subsidies for modern techniques by government
- ✓ Selective, cross breeding on scientific lines for better animals
- ✓ Better feed for stronger and healthier animal
- ✓ More grazing land
- ✓ Control of disease
- ✓ Vaccination to improve health
- ✓ More medicines/vets to treat animals
- ✓ Better hygiene, care
- ✓ Mechanization.